

AtmosMesh bench station – rev A

Confirmed bench wiring (operator, 2026-08-14):
OLED 0x3C SDA = GPIO5 SCL = GPIO4 VCC = 3V3
GY-BMP280 SDA = GPIO21 SCL = GPIO19 VCC = 3V3
CSB = 3V3 (I2C) SDO = GND (addr 0x76)
AM2302/DHT22 DATA = GPIO18 (10k pull-up R1) VDD = 3V3
BEEPER SIG GPIO25 PIR D-SUN SIG GPIO33
VEML7700 lux SDA = GPIO21 SCL = GPIO19 (shared bus, 0x10)
GPIO2 LEFT FREE – download strap, do not load

Future 5 V sensors (connectors only, off-board enclosed PSU):
SDS011 5 V supply, 3.3 V TTL UART on GPIO16/GPIO17
MQ135 5 V heater, analog out through R2/R3 to GPIO34

RULES

1. No 5 V on any ESP32 GPIO or on the 3V3 rail. Common GND only.
2. 3V3 comes from the DevKit onboard LDO. Do not back-feed it.
3. JP1 (5V → DevKit VIN) stays OPEN on the bench. USB powers the DevKit.
4. No mains on this board. The AC/DC module is off-board and enclosed.

U1 PIN ORDER IS PROVISIONAL. The DevKit silkscreen has not been photographed yet. Verify every pin against the physical board before fabrication; J7 is the flying-lead fallback if the order differs.

